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Changes in Physical Status of Young Women in Response to Exercise Training

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Abstract

Fitness-related benefits result in essential changes in physical fitness level in young women. This study was conducted with fitness centre clients (35 women with an average age of 26.4 ± 2.2 years). The results of the experiment made it possible to evaluate the short-term response of the body to the training loads of different intensities in different blocks of the training session. The following six main blocks were distinguished in the structure of the exercise sessions including: aerobic, dance, variable, corrective, warm-up, and relaxation; and two additional ones. The aerobic and dance blocks were characterized by the highest intensity: HRav was $75.6 \pm 1.71\%$ of HRmax, VO₂ was $63.5 \pm 1.64\%$ of VO₂max, and energy expenditure (EE) was ranged from 465.28 to 524.36 kcal/hour. The relaxation block had the lowest intensity: HR was $57.31 \pm 0.56\%$ of HRmax, VO₂ was $34.2 \pm 6.25\%$ of VO₂max, and EE was of 252.6 ± 49.24 kcal/h. We found that the percentage contribution of fats to energy production was higher in the warm-up (78.24%), relaxation (68.93%), and corrective (42.65%) blocks. The largest percentage of CHO utilization for energy metabolism was found in additional (92.8%), variable (91.82%), and dance (86.54%) blocks. The data obtained allowed to experimentally substantiate and methodologically structure a number of fitness means used in exercise sessions, by dividing them into specialized blocks that made it possible to selectively effect on different components of the physical condition.

Keywords: physical fitness level, acute effect, fitness training, mature age women

Introduction

Nowadays, there are a wide variety of physical activity forms. Therefore, the number of scientific studies in area of fitness-related benefits are increasing. Investigators in different fields of sports science and medicine interact with each other, and professionals strive to use the latest scientific advances and successful finds of fitness and health promotion practitioners.

Various type of fitness training such as aerobics, dance aerobics, etc. are the most popular among women in the health promotion practice (Blagii, & Lysakova; 2013; Yarmak et al., 2018).

The physical fitness of women and approaches to improve it through the use of various forms of health-promoting activities have been studied by many authors (Gubareva, 2001; Baechle, & Earle, 2008; Ivaschenko, 2008; Bulatova, Andrieieva, & Blagii

2017; Drozdovska, Andrieieva, Yarmak, & Blagii, 2020; Hakman et al., 2020). At present, the effects of rhythmic gymnastics training are described quite well, methodological characteristics of step aerobics programs are identified (Gubareva, & Levitsky, 2000). There has been a tendency to systematize and generalize the accumulated experience (Howley, & Franks, 2003; Baechle, & Earle, 2008; Bulatova et al., 2017). However, the contradictions in the methodology of various forms of fitness classes do not allow the integrated use of their capabilities.

Unfortunately, the use of new forms of fitness is not always sufficiently substantiated. Furthermore, various instructor training courses and fitness conventions in their methodological sections do not take into account the fundamental principles of designing exercise classes using various forms of fitness



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training. The continual change of emerging fitness activities creates a problem for clients who have to make the appropriate choice of individual means of health improvement. While trying to keep up with the trends of the time, fitness instructors have to change their workout routine frequently. Obviously, there is a need for the rational use of all the capabilities inherent in various methods.

The aim of this study was to investigate the acute effects of fitness training on physical fitness variables in active young women.

Methods

The study involved 35 women aged from 21 to 30 with an average age of 26.4 ± 2.2 years. The women were found to have no health abnormalities during the medical examination. The selection criteria included women's willingness to participate in the experiment, participation in health fitness activities for at least one year, and high levels of physical activity and physical fitness. The exclusion criteria included pregnancy, acute respiratory diseases, and chronic diseases in the acute phase. The study was conducted in accordance with the ethical principles of the World Medical Association Declaration of Helsinki. The study was conducted at the National University of Ukraine on Physical Education and Sport (Kyiv, Ukraine). Ethical approval of research protocol was obtained from the Ethics Commission of the NUUPES (No. 2 on 16.12.2020). All the participants gave their written informed consent to the personal data processing, as well as to further processing of the obtained data for scientific purposes.

Body composition and physiological parameters measurements were carried out with a focus on identifying characteristics of the acute effects of physical exercise classes designed using different methodologies. The study of the effects of various modes of exercise training on the body in terms of heart rate (HR) changes (using the "Polar" heart rate monitor) included comparisons of different subjects participating in the same model sessions and functional responses of individual subjects during different model sessions. To measure physiological parameters, the wearable metabolic system for assessing cardiorespiratory function Cosmed K5 was used. The following body parameters were recorded: oxygen uptake (VO_2

in $\text{mL} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$); HR (bpm); carbon dioxide output (VCO_2 , $\text{mL} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$); lactate level (La, $\text{mmol} \cdot \text{L}^{-1}$); fat utilization as percentage of the total energy production (FAT%); carbohydrates (CHO) utilization as percentage of the total energy production (CHO%); energy expenditure (EE) per minute (EE_m , $\text{kcal} \cdot \text{min}^{-1}$); EE per hour (EE_h , $\text{kcal} \cdot \text{hr}^{-1}$); daily EE per kilogram of body weight (EE/kg , $\text{kcal} \cdot \text{kg}^{-1} \cdot \text{hr}^{-1}$); daily EE per body surface area (EE/BSA , $\text{kcal} \cdot \text{m}^{-2} \cdot \text{hr}^{-1}$). METs are multiples of basal metabolic rate, 1 MET is equal to approximately $3.5 \text{ mL} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$.

A group of the subjects ($n=35$) performed two different model exercise sessions with a fitness instructor. In addition to the video recording of the session, HR changes and gas exchange variables were recorded every 15 seconds for every subject. Then, the following indicators were calculated: EE_m , EE_h , CHO and fat utilization as percentage of the total energy production (CHO% and FAT%). Blood lactate concentration (HLA, $\text{mmol} \cdot \text{L}^{-1}$) was measured using an LP-400 biochemical analyzer (Dr.Lange, Germany). Blood HLA values were measured at the beginning, middle, and end of the workout. Blood samples were obtained by the staff at the NUUPES Research Institute.

The study was focused on assessing the training effect of the model exercises; comparing the sessions #1 and #2; evaluating the relative effectiveness of the blocks composing the sessions; and assessing the individual response of the subjects to the standard training load during the sessions.

Statistical analysis of the data was performed using a software package Statistica 10.0 (StatSoft, Inc., USA). With an average sample size ($30 \leq n \leq 100$), the normality of the distribution was tested by several methods including the Shapiro-Wilk test as well as asymmetry and excess kurtosis. One-dimensional statistical analysis, which involved calculating the mean and standard deviation (SD), was applied in the research. The parametric Student t-test was used to determine the significance of the difference between the groups participated in two different exercise sessions. The significance for all statistical tests was set at $p \leq 0.05$.

Results

During the study, we assessed the acute effects of two model exercise sessions with different fitness goals. We have developed a generalized structure of health-enhancing exer-

Table 1. Characteristics of exercise load in the blocks of training session ($\bar{x} \pm \text{SD}$)

T, min	V	I	AE			HRav		HR range
			D	S	Str	bpm	% HRmax	HRmin - HRmax
Warm-up block								
7.6±1.46	171.1±21.31	0.304±0.04	67.4±4.13	11.1±1.62	21.2±1.13	119.2±3.91	62.2±1.32	98-135
Aerobic block								
16.4±0.42	504.3±21.16	0,85±0.01	100	0.0	0.0	155.7±6.14	80.8±1.04	142-168
Dance block								
9.1±1.03	464.2±60.1	0.76±0.01	99.0	1.0	0.0	141.7±3.04	74.2±0.91	132-157
Preventive block								
8.4±1.06	199.5±96.2	0.31±0.12	54.0±2.11	42.5±10.42	3.5±1.14	103.2±4.12	53.6±4.32	93-116
Relaxation block								
6.6±0.68	17.1±12.4	0.03±0.01	30.0±9.61	15.0±2.71	55.0±44.6	97.0±4.62	50.0±2.11	87-105
Additional block								
7.1±0.32	273.1±41.1	0.45±0.11	85.5±4.12	14.5±1.40	0.0	131.0±9.0	68.0±5.4	91-145

Legend: T – time; V – volume of training elements; I – intensity, elements per second; AE – amount of exercise, % of total time; D – dynamic, S – static, and Str – stretching exercise; HRav – HR average.

cise sessions on the basis of the block principle of design. The blocks are characterized by different fitness goals, which together cover the full range of health-enhancing fitness tasks and take into account the preferences of participants. The following blocks were distinguished in the structure of the ex-

ercise sessions including warm-up, aerobic, dance, corrective, preventive, relaxation, additional, and variable. Pedagogical and medical biological characteristics of the exercise blocks (Table 1) demonstrated their specialized effects on the body of the women.

Table 2. Parameters of the model sessions in comparison with the other forms of exercise training (n=35), ($\bar{x}$ ±SD)

Type of activity	VO ₂ (mL·kg ⁻¹ ·min ⁻¹)	% VO ₂ max	HR, bpm	% HRmax	Fat %	CHO%	EEh, kcal·hr ⁻¹
Spinning	33.7±2.18	67.4±4.28	164.7±11.34	85.0±6.81	8.1±2.04	92.1±6.18	529.9±64.18
Fitball aerobics	32.9±3.24	65.2±3.29	163.5±14.26	84.0±5.33	30.2±2.32	70.2±6.34	516.7±55.23
Slide aerobics	32.4±2.43	64.7±5.14	162.4±10.55	84.0±7.18	36.5±3.09	63.4±5.78	511.2±46.61
Step aerobics	30.5±1.18	61.6±4.38	146.3±11.87	76.0±4.39	46.1±3.19	54.1±4.19	475.3±58.21
Model session 1							
Aerobic block	27.9±2.14	66.6±5.22	145.2±12.51	75.0±6.89	33.2±2.76	67.2±6.27	525.8±44.16
Corrective block	21.3±2.21	48.6±3.67	143.1±13.32	74.0±6.92	40.3±3.84	59.8±3.61	373.5±41.27
Model session 2							
Aerobic block	27.3±3.45	64.9±6.58	141.7±12.98	72.0±7.11	30.1±1.54	70.1±5.27	498.2±52.17
Relaxation block	16.0±2.54	39.9±3.54	115.1±10.47	57.0±5.07	34.2±2.27	66.1±5.84	286.8±48.32

The obtained results were compared with the available data on the response to the exercise load in various aerobics workouts (Table 2).

The experimental sessions, which lasted for 45 min, consisted of 6 blocks: 1. Warm up (introductory block, 5 min). 2. Functional training (aerobics and dance blocks, 15 min). 3. Recovery (restorative block, 5 min). 4. Training of the musculoskeletal system (corrective block in the session #1 and preventive block in the session #2, 10 min). 5. The development of creative skills and motor abilities (variable block in the session #1 and additional block in the session #2, 5 min). 6. Relaxation (cool down block, 5 min). The session #2 differed from the session #1 in two blocks: the preventive block was used instead of the corrective, and the additional block was used instead of the variable one.

Relaxation block was used twice in both sessions. This block included walking, breathing exercises, and stretching in a standing position.

For the second time, the relaxation block was used in the final parts of the sessions aimed at reducing fatigue and psycho-emotional stress and consisted of stretching exercises, yoga postures as well as elements of autogenic training and massage.

The results of the study revealed that the experimental sessions #1 and #2 provided an equal exercise load. Significant differences between the sessions ($p < 0.05$) were found only in HR values.

HR_{av} in the session #1 was 135.4±8.2 bpm or 69.7±2.8% of HR_{max}. VO₂ was 22.1±3.1 mL·kg⁻¹·min⁻¹ or 49.5±3.7% of VO₂max, and EE was 379.9±98.76 kcal·hr⁻¹. About 56.7% (more than 25 min) of the total duration of sessions was performed at HR of 149.0±5.4 bpm (75.9±2.1% of HR_{max}) and VO₂ of 24.9±0.12 mL·kg⁻¹·min⁻¹ (61.2±3.1% of VO₂max) that confirms the sufficient load on the cardiovascular system.

The HR_{av} in the session #2 was 129.2±10.4 bpm (67.12±5.04% of HR_{max}); VO₂ was 23.01±2.18 mL·kg⁻¹·min⁻¹, 6.01 METs respectively (51.4±8.51% of VO₂max).

The mixed aerobic-anaerobic nature of the exercise load was indicated by the lactate levels: 3.67±1.21 mmol·L⁻¹ (in the middle of the session #1); 4.55±1.79 mmol·L⁻¹ (at the end of

the session #1); 3.75±0.22 mmol·L⁻¹ (in the middle of the session #2); and 3.15±0.58 mmol·L⁻¹ (at the end of the session #2).

The experimental sessions were less intense than spinning, fitball, and slide aerobics sessions and had an intensity similar to that of step aerobics sessions (Table 1).

During the experiment, 8 blocks with different fitness goals were examined, and the results are presented in table 2. When comparing the same blocks in the sessions #1 and #2, no statistically significant differences were found ($p > 0.05$). This indicates the sessions had a well-balanced plan and division of exercises into the blocks. Exercises of the same block caused a similar physiological response with close values of HR, VO₂, EE, and the proportion of CHO and fat utilization in energy metabolism. Exercises of different blocks differed from each other.

The difference in the goals and specialized impact of the studied blocks was confirmed by the study of their effects. The intensity of the model exercise sessions developed using block design was sufficient to stimulate aerobic metabolism and corresponded to HR of 68.6±1.32% of HR_{max}, VO₂ of 51.4±1.46% VO₂max, EE of 383.5±61.12 kcal·hr⁻¹, and a metabolic rate of 6.2 MET. Rational alteration of different blocks in exercise sessions increased the effectiveness of addressing specific objectives of fitness training.

The intensity of the exercise load in the dance block was close to anaerobic threshold: HR ranged from 145.0 to 150.0 bpm (74.85-77.61% of HR_{max}), VO₂ varied from 21.77 to 26.12 mL·kg⁻¹·min⁻¹ (56.84-62.77% of VO₂max). The aerobic block was performed primarily in aerobic and partially in mixed mode: HR was from 141.0 to 146.0 bpm (73.21-75.82% of HR_{max}), VO₂ was from 27.38-27.93 mL·kg⁻¹·min⁻¹ (65.12-66.73% of VO₂max).

The greatest EE was during the aerobic block: 524.36±19.32 kcal·hr⁻¹. We found that the percentage contribution of fats to energy production was higher in the introductory (78.24%), relaxation (68.93%), and corrective (42.65%) blocks. The largest percentage of CHO utilization for energy metabolism was recorded in additional (92.8%), variable (91.82%), and dance (86.54%) blocks. In these blocks the exercise training mode was closer to the anaerobic and intensity was higher.

Table 3. Functional indicators and energy expenditures of middle-aged women in model sessions (n=35), ($\bar{x} \pm SD$)

Block	VO ₂	HR	%VO ₂ max	%HR max	EEm	EEh	EEkg	FATg	CHOg	FAT%	CHO%	FAT mg	CHO mg	MET
					Kcal/ min	Kcal/h	Kcal/ Kg/day	g/day	g/day	%	%	mg/min	mg/min	
Model session 1														
1	18.0±	124.0±	42.8±	63.7±	5.52±	326.2±	127.4±	198.6±	1429.4±	24.1±	75.1±	137.7±	993.3±	5.2±
	1.24	8.53	8.26	7.16	0.87	93.26	28.33	21.18	93.37	6.81	4.53	25.16	95.21	0.36
2	27.9±	146.0±	66.7±	75.8±	8.7±	524.4±	203.3±	431.3±	2039.5±	33.5±	67.1±	301.2±	1415.8±	8.0±
	1.33	9.38	6.82	5.39	1.17	76.31	47.29	75.12	98.14	4.11	5.22	64.91	98.82	0.47
3	26.1±	150.0±	62.8±	77.6±	7.8±	465.3±	184.5±	422.9±	1734.7±	33.6±	67.0±	291.2±	1204.7±	7.6±
	2.18	9.71	3.85	4.82	5.73	74.28	26.72	89.56	99.24	2.11	3.42	43.92	97.14	1.48
4	13.5±	110.0±	30.2±	56.2±	3.6±	217.4±	84.3±	394.3±	356.9±	68.9±	30.6±	273.0±	247.2±	3.8±
	1.42	7.33	3.17	4.67	1.07	76.53	10.25	85.29	83.76	7.32	2.69	94.75	83.91	0.95
5	21.3±	141.0±	47.9±	72.1±	6.2±	374.1±	143.5±	483.5±	103.8 ±	42.6±	60.0±	333.3±	732.1±	6.1±
	2.35	8.95	5.16	8.41	1.05	95.34	28.48	75.31	31.06	4.11	5.04	63.27	58.37	1.07
6	22.0±	139.0±	51.7±	71.7±	6.6±	394.9±	154.8±	281.5±	1661.4±	27.2±	73.3±	193.7±	1153.5±	6.4±
	2.87	9.36	4.68	6.34	0.84	73.91	18.24	48.94	75.48	1.49	8.55	18.93	82.79	1.02
Model session 2														
1	19.6±	117.0±	48.9±	60.72±	5.8±	339.7±	131.1±	736.7±	303.3±	78.2±	23.2±	513.1±	213.4±	5.6±
	1.31	9.61	4.28	5.18	1.09	58.29	19.37	69.93	45.28	7.33	4.86	48.83	48.49	1.05
2	27.3±	141.0±	65.1±	73.2±	8.4±	495.8±	192.4±	393.8±	1968.5±	30.2±	70.4±	274.7±	1371.7±	7.8±
	1.16	9.28	5.16	7.84	1.36	68.74	22.87	75.72	201.46	1.54	5.90	54.89	98.65	0.97
3	21.7±	145.0±	56.8±	74.8±	7.1±	426.7±	163.5±	152.5±	2132.2±	13.0±	86.5±	104.6±	1472.6±	6.5±
	2.18	9.54	8.36	5.88	1.09	77.12	18.26	14.53	204.21	0.78	6.43	11.21	45.28	0.65
4	15.9±	113.0±	41.2±	58.1±	4.8±	285.7±	113.4±	263.2±	1061.4±	34.2±	65.7±	182.4±	734.6±	4.6±
	1.13	8.59	4.78	5.21	0.41	48.91	22.19	18.28	97.14	2.18	3.75	12.67	42.46	0.61
5	18.2±	127.0±	44.8±	66.2±	5.4±	323.1±	125.6±	194.6±	1431.5±	21.2±	78.9±	135.6±	992.5±	5.1±
	1.87	9.45	2.18	7.03	0.88	87.12	14.59	20.84	86.21	2.65	4.13	18.59	51.13	0.82
6	23.2±	138.0±	59.2±	71.1±	7.3±	434.2±	161.6±	112.6±	2260.2±	7.4±	91.8±	77.9±	1568.3±	6.6±
	2.17	8.49	3.11	6.38	1.25	56.74	12.45	10.21	215.67	1.82	8.39	4.63	37.81	1.22

Legend: 1 – Warm-up block, 2 – Aerobic block, 3 – Dance block, 4 – Preventive block, 5 – Relaxation block, 6 – Additional block

During the experiment, blocks with similar exercise intensity were identified that can replace each other in the session (table 3).

Such interchangeable blocks include:

1) aerobic block at HRav of 146.0 bpm (75.82% of HRmax) and dance block at HRav of 150.0 bpm (77.61% of HRmax); these blocks have the same objective of functional training, but through the use of different means;

2) corrective block at VO₂ of 21.35 mL·kg⁻¹·min⁻¹ (47.93% of VO₂max) and preventive block at VO₂ of 18.21 mL·kg⁻¹·min⁻¹ (44.76% of VO₂max) often consist of the same exercises, which can be focused on addressing different objectives. For example, the exercises for the abdominal muscles from the corrective block may be the same as the exercises for the muscles of the pelvic floor from the preventive block;

3) warm-up block at HRav of 124.0 bpm (63.73% of HRmax) and relaxation block at HRav of 110.0 bpm (56.24% of HRmax);

4) additional block at VO₂ of 23.26 mL·kg⁻¹·min⁻¹ (59.21% of VO₂max) and variable block at VO₂ of 22.01 mL·kg⁻¹·min⁻¹ (51.72% of VO₂max).

The results of this study showed that the same exercises and the same exercise sessions may affect the body of the subjects in different ways.

Thus, it was found that the body can utilize different sources for energy production depending on the initial level of physical fitness of the subjects (with a volitional effort to

continue the exercise).

The higher the intensity of exercise, the lower the proportion of fat contribution to energy production. In addition, the lower the level of physical condition, the lower should be the intensity of exercise. This confirms the need for taking into account the level of physical condition when completing groups and programming self-training workouts.

The differences in utilization of energy sources that was identified experimentally must be taken into account when designing exercise sessions. The transition from low-intensity to high-intensity workouts is possible only on the basis of increasing the level of physical condition.

The lactate threshold is defined as the point when blood lactate begins to substantially accumulate above resting concentrations during exercise of increasing intensity. Increased exercise load results in a faster accumulation of lactate. Table 4 presents the blood lactate concentration data. During the exercise session, the intensity of exercise can exceed the anaerobic threshold and the body switches to aerobic-anaerobic energy metabolism. For example, during spinning classes, blood lactate reaches 6.07±2.04 mmol/L in the middle of the session and 5.30±1.79 mmol/L at the end of the session. Activities during fitball, slide aerobics, and step aerobics classes mainly depend on aerobic production of energy.

Based on the obtained average lactate measurements (3.15±0.58; 3.67±1.21; 3.75±0.22; and 4.55±1.79 mol/L), the model sessions were performed at intensity levels below the

Table 4. The changes in blood lactate during various forms of exercise sessions (n=35)

Forms of sessions	Blood lactate level ($\bar{x} \pm SD$, mmol/L) ($x_{min} \div x_{max}$)		
	Baseline (at the beginning of a session)	Intermediate (in the middle of a session)	Final (at the end of a session)
Spinning	0.93±0.24	6.07±2.04 (3.25÷8.03)	5.30±1.79 (3.19÷7.55)
Fitball	1.26±0.15	2.39±1.04 (1.46÷3.72)	2.07±0.82 (1.36÷3.11)
Slide aerobics	1.94±0.10	2.41±0.30 (1.97÷2.58)	2.45±0.46 (1.83÷2.75)
Step aerobics	1.52±0.11	1.84±0.11 (1.75÷1.97)	2.40±0.14 (2.21÷2.61)
Model session #1	1.14±0.29	3.67±1.21 (1.81÷6.52)	4.55±1.79 (2.64÷8.73)
Model session #2	1.03±0.22	3.75±0.22 (2.86÷4.68)	3.15±0.58 (2.51÷4.12)

anaerobic threshold. However, an analysis of individual values of lactate level showed that in the middle of the session a number of the subjects switched to the anaerobic mechanism of energy production due to their level of physical condition. This indicated a mismatch between the exercise load and functional capabilities of the subjects.

When assessing the proportion of substrates utilized for energy production during the warm-up (session #1), preventive, and recreational (session #2) blocks, measurement artifacts were identified, namely, an overestimated carbohydrate utilization (CHO%). These blocks included moderate and low intensity exercises. When changing the exercise intensity, for example, by changing from a standing starting position to a sitting or lying position, or when finishing dynamic exercises, breathing typically is enhanced that is accompanied by the release of CO₂. The so-called 'false increased CHO%' does not occur due to the increased metabolism of the energy substrates, but due to the buffer activity of the device, i.e. its inertia. This results in the error in the estimation of CHO%.

Discussion

The range of fitness classes and the objectives which they address is very wide. This fact, as well as the availability of fitness equipment, the opportunity for year-round training, and low-cost requirements make fitness training a preferred form of health-promoting physical activity.

In health-promoting fitness training, two directions can be distinguished: strength training and aerobic training (dance fitness). The first is focused mainly on improving body shape, while the focus of the second is beauty and originality of movements and movement compositions. Their well-thought-out combination is the most successful.

There is a lot of publications on the influence of health-promoting exercise on the body (Baechle, & Earle, 2008; Bulatova et al., 2017; Drozdovska et al., 2020; Galan, Andrieieva, & Yarmak, 2019a; Gubareva, 2001; Hakman et al., 2020). There have been extensive medical and biological studies of the functional response of the body to exercise load (Powers, & Howley, 2007). A number of researches covering individual health-promoting techniques and their options have been completed (Blagii, & Lysakova, 2013; Galan, Andrieieva, Yarmak, & Shestobuz, 2019b; Nagorna, & Andrieieva, 2019). The general principles for designing health-promoting exercise training programs have been developed (Ivashchenko et al., 2008). Nevertheless, the technology for designing exercise sessions has not been sufficiently developed, and the effectiveness of the influence of aerobic and strength training on the body has not been fully studied. The contradictions between the level of knowledge reached in the field of health-promoting fitness training and

the development of technology for designing exercise workouts create obstacles to the integrated use of these types of fitness training among the population groups which differed in the health-promoting exercise interests and in the levels of physical condition. An analysis of different forms of fitness classes suggests that there are many methodological approaches to design and development of physical exercise programs, each of which has its own advantages and disadvantages. Each of them has a right to exist, but there is no one universal approach suitable for addressing the whole range of various objectives of health promotion plans. Depending on the goals, exercise sessions can address pedagogical, health-promoting, aesthetic, developmental, rehabilitation, autogenic training, and other objectives (Zoeller, 2007; Ciccolo, Carr, Krupel, & Longval, 2010; Kashuba et al., 2019). The diverse objectives of the exercise sessions are due to the numerous and diverse motives and interests of the involved women.

Our main findings are that the rational alteration of different blocks in exercise sessions can increase the effectiveness of addressing specific objectives of health-promoting exercise training. The intensity of the model exercise sessions developed using block design was sufficient to stimulate aerobic metabolism and corresponded to HR of 68.9±2.13% of HR_{max}, VO₂ of 52.1±2.11% VO_{2max}, EE of 386.0±62.0 kcal·hr⁻¹, and a metabolic rate of 6.2 MET. Our results confirmed the data reported on the effectiveness of the influence of strength and aerobic training on the functional condition of the body in young women.

The data obtained allowed us to identify, experimentally substantiate, and methodologically structure a set of training means used in exercise sessions, by dividing them into specialized blocks that made it possible to selectively affect different components of the physical condition. Rational alteration of different exercise blocks in the workouts allows to program exercise sessions with different goals and intensity depending on the age, fitness level, and interests of women. Our study confirmed the available data that the block structure is the most rational choice for an integrated exercise session design. This research extended the data on the effectiveness of the influence of integrated health-promoting exercise sessions with different physical condition goals in young women. Furthermore, new data were obtained to experimentally substantiate the use of specialized blocks that make it possible to selectively affect various components of the physical condition of young women.

Conclusions

The data obtained allowed to experimentally substantiate and methodologically structure a number of fitness means

used in exercise sessions, by dividing them into specialized blocks that made it possible to selectively effect on different components of the physical condition. The results of assessing

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Conflict of Interest

The authors declare that there are no conflicts of interest.

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